

Responsible Software - CheatSheet

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INTRO. Responsible Software. Software engineered in a responsible (*ethical*) way.

- 1. With a goal to do good.
- 2. While preventing avoidable negative impacts.
- 3. Taking people, other systems, social structures and our planet into account.

Active Responsibility. anticipating/preventing negative impacts before occurring.

Ethics. Principles/standards that guide human behavior in various contexts.

Metaethics. "what does it mean for an action to be right?"

Normative. "how should people act in general (moral principles)?"

Applied. (Our approach) "how can i act ethically in a real-world situation?"

Technical decisions can have *ethical* implications.

Ethical Issue. any situation that may compromise respect of ethical value/principle.

Ethical Agency. capacity to take actions that align with ethical values and principles.

Ethical Sensitivity. ability to

- recognize ethical issues (awareness)
- identify their impact: who, what

Ethical Decision Making.

Recognize(knowledge+experience+habits)→**Evaluate**(cognitive+reason+ffective)→**Act**.

Ethical Lenses.

- 1. **Desirability.** The product corresponds to what people want.
- 2. **Viability.** The product generates profits for a business.
- 3. **Feasibility.** The product can be created with existing or new technology.

Fourth criterion (Isaac et al.). Add **Ethicality:** the product should also be ethical.

Ethical Dilemmas. decision between multiple alternatives, all conflict with ethical values/principles. *None of the options override the other, wrong-wrong choice.*

Moral side. **Cognitive side (Normative Ethics).**

- **Utilitarianism** Consequences: choose the greatest good for greatest number
- **Deontology** Action: do the right thing according to universal principles
- **Virtue** Person: act as a virtuous person
- **Care** Relationships: mutual responsibility and care for each other

Trolley problem. thought experiment, **choice:** do nothing → runaway trolley kill people, or intervening → fewer people die. tests which moral principles justify action vs. inaction.

Ethical Decision Making Frameworks.

- 1. Identify the ethical issues
- 2. Get the facts
- 3. Evaluate alternative actions with *several ethical theories*
- 4. Choose an option and "test" it
- 5. Implement your decision and reflect on the outcome.

Stakeholder Individuals, groups, organizations, institutions, and societies that might reasonably be affected by technology.

Direct Stakeholder. *in contact* with the product (*End-users, Admins, Contractors, Developers*).

Indirect Stakeholder. affected through *consequences of the usage* of the software (*Environment, Government, Public(s), NGOs, Friends, Neighborhood*).

Stakeholder analysis. Identifying every entity that can be affected by software positively/negatively.

- 1. Brainstorm to find all stakeholders possible using questions.
- 2. Use Direct vs Indirect Chart to find new stakeholders.

Additional Methodologies: *Ethix. add unintended stakeholders, Mitchell. add attributes to stakeholders, SAS. use different chart with scale for impact from product and influence on the product.*

Ethical Speculation. How to think, not what to think - anticipate potential impacts on stakeholders.

Create a fictional story in two parts.

- 1. Check pessimistic outcomes *Black Mirror*.
- 2. Positive resolutions and future scenarios.